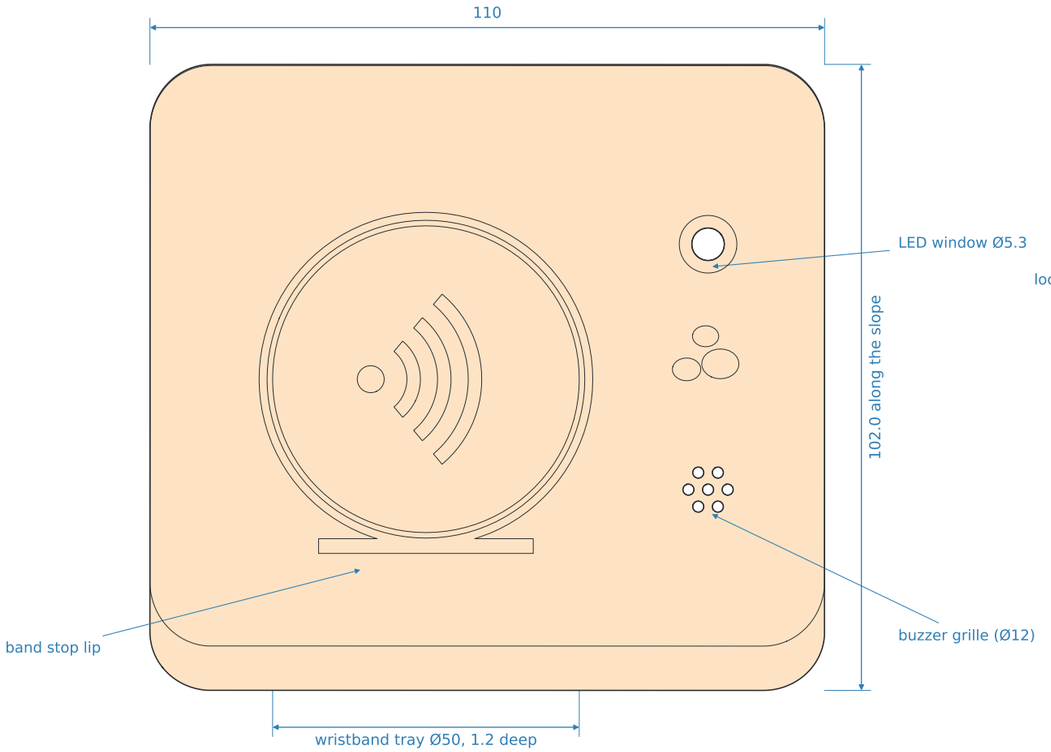
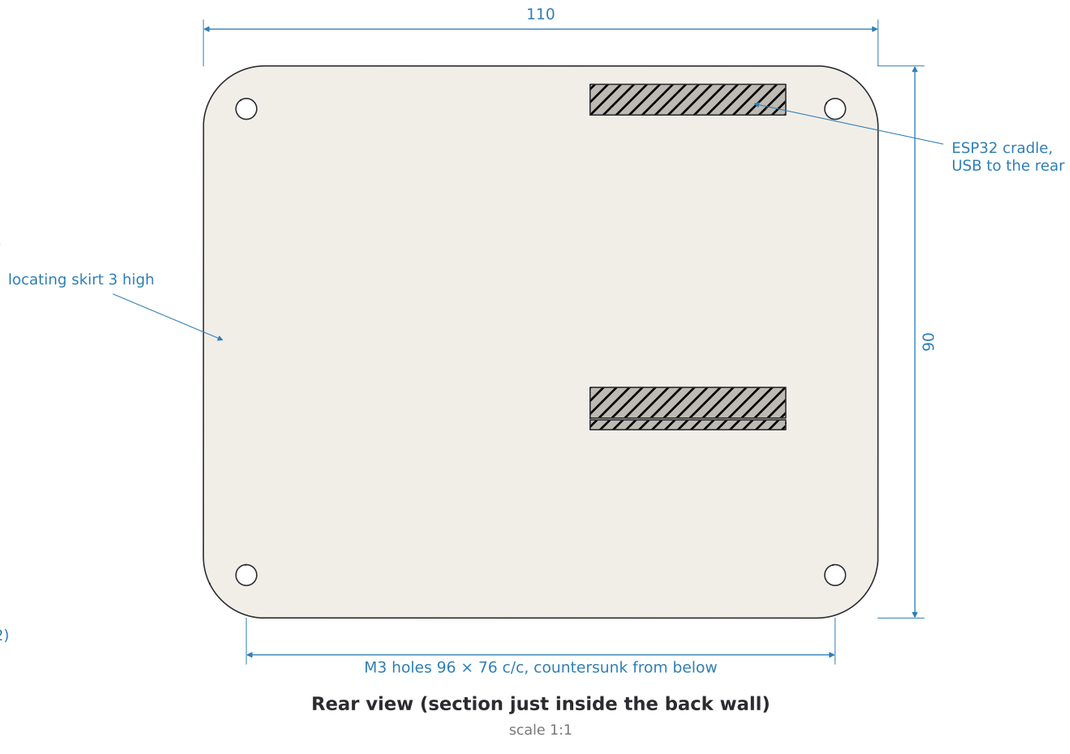
Shell, view square to the sloped top

scale 1:1



Bottom plate, section at 4 mm

scale 1:1

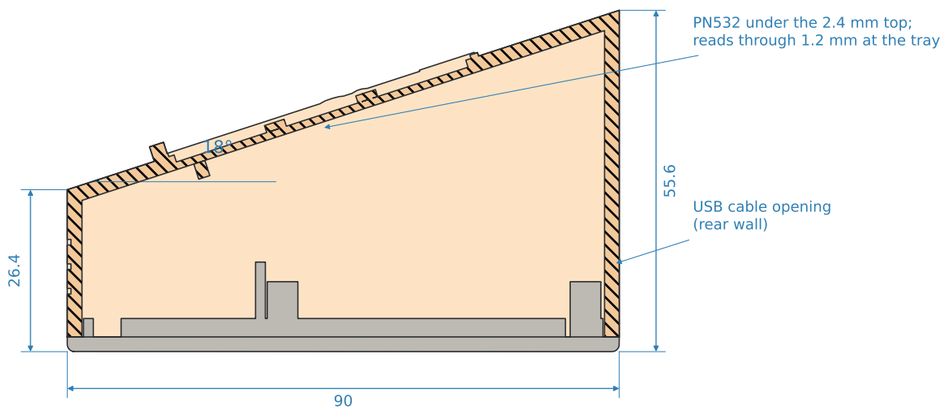


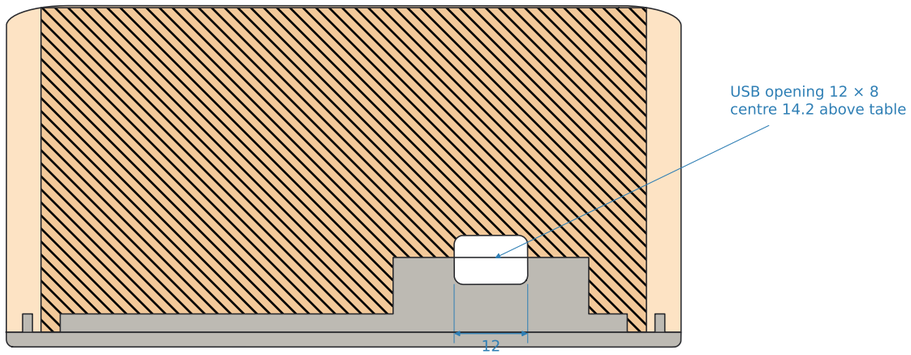
Rear view (section just inside the back wall)

scale 1:1

Side view + section A-A (through the tray)

scale 1:1





Components (sizes assumed, check yours)

ESP32 DevKitC V4	54.4 × 27.9 mm, pins down
PN532 V3 board	42.7 × 40.4 mm, I2C (SDA 21, SCL 22)
Buzzer	Ø12 × 9.5 mm, GPIO 4
LED	5 mm, GPIO 2
Screws	M3 × 10 self-tapping, countersunk ×4
Feet	Ø10 rubber bumpers ×4

Printing and assembly

Shell: tangerine PLA/PETG. The STL lies on its sloped top; supports on (build plate only).
Plate: basalt grey, flat side down.
0.2 mm layers, 3 walls, 15–20 % infill.
PN532: foam tape in the frame under the tray.

Admin desk unit

Sheet 1/2

Sloped shell + bottom plate for ESP32 + PN532

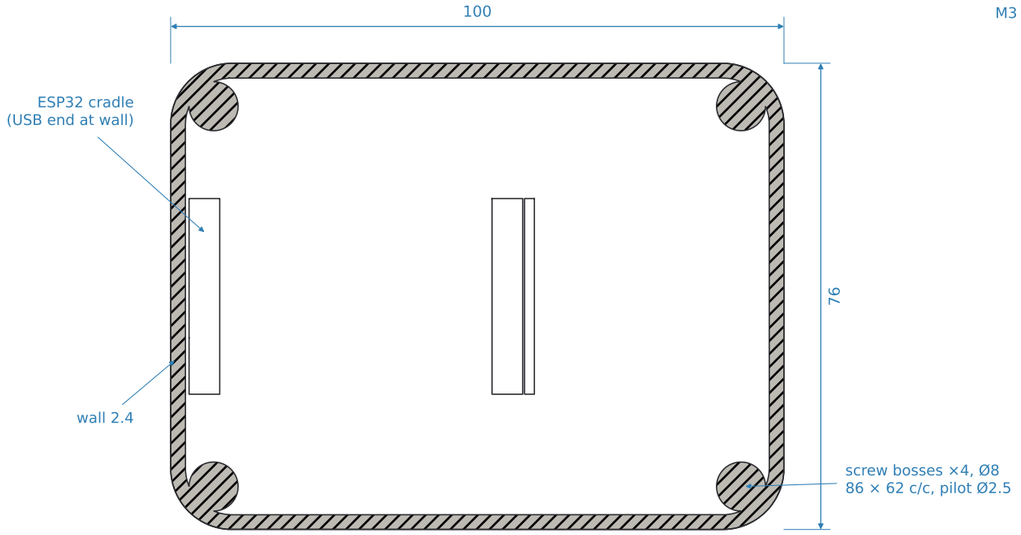
Units: mm

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A3 · print at 100 %

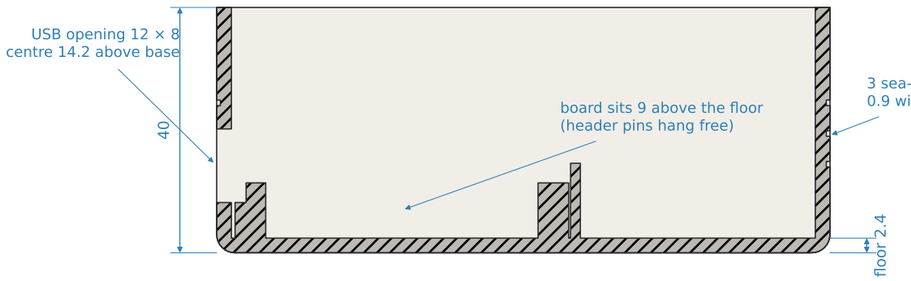
Base, section at 8 mm (top view)

scale 1:1



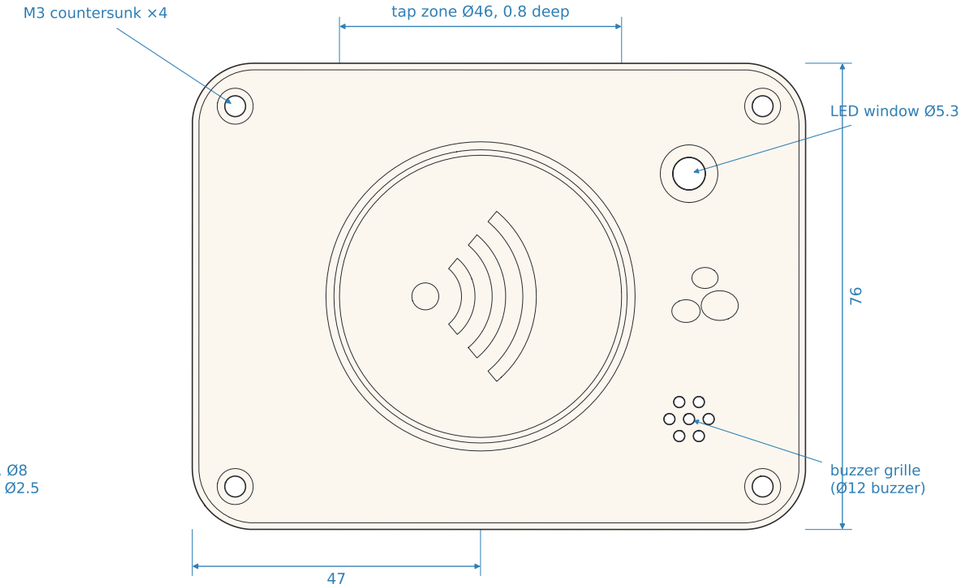
Base, section B-B (front)

scale 1:1



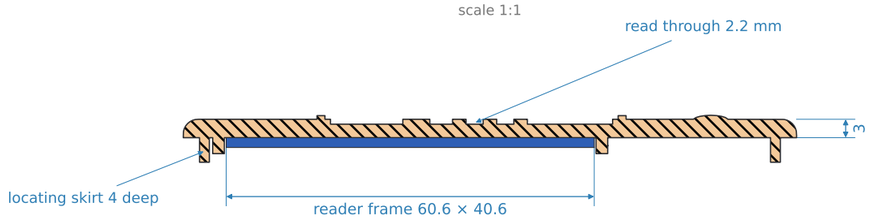
Lid, top view

scale 1:1



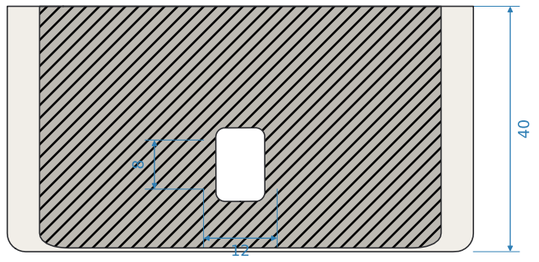
Lid, section C-C with RC522

scale 1:1



Base, left side (USB end)

scale 1:1



Components (sizes assumed, check yours)

ESP32 DevKitC V4	54.4 × 27.9 mm, pins down
RC522 board	60 × 40 mm (food / place / activity)
PN532 V3 board	42.7 × 40.4 mm (admin desk)
Buzzer	Ø12 × 9.5 mm, GPIO 4
LED	5 mm, GPIO 2
Screws	M3 × 8 self-tapping, countersunk ×4
Feet	Ø10 rubber bumpers ×4

Printing and assembly

Base PLA/PETG basalt grey, open side up, no supports (USB opening bridges 12 mm).
Lid PLA/PETG cream, flat top on the bed (STL is already flipped), supports off.
0.2 mm layers, 3 walls, 15–20 % infill.
Reader: stick the RC522 in the frame with foam tape, antenna end under the tap circle; same frame takes the PN532 for the admin desk.

Check-in station

Sheet 2/2

Food / place / activity · base + lid for ESP32 + RC522

Units: mm

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A3 · print at 100 %